Relaytic-AML: A Local-First Agentic Evaluation Lab for Financial-Crime Machine Learning

TODO_EVIDENCE[author_metadata]
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Abstract

Anti-money laundering (AML) machine-learning results are hard to trust when privacy posture, temporal validity, graph provenance, leakage controls, review capacity, and public claims are reported as separate objects. Relaytic-AML introduces a local-first artifact graph, cooperating role-scoped agents, evidence cells, and claim gates that bind each result to its data, split, command, budget, operating point, and allowed interpretation. In the current public evidence pack, PaySim reports supporting synthetic temporal-fraud PR-AUC 0.638773, Elliptic reports supporting temporal graph-feature PR-AUC 0.668756, and Elliptic2 is retained as modern-context evidence with official-partition PR-AUC 0.94324 +/- 0.000882, below the recorded RevClassifyDS reference of 0.974. The contribution is a reproducible evaluation-lab architecture for governed AML experimentation, not a detector-superiority claim.

1 Introduction

AML modeling is a rare-event, high-stakes evaluation problem. The input may be a stream of mobile-money transfers, card or account events, wire messages, customer profiles, or blockchain transactions. Suspicion rarely lives in one row. It often appears as a pattern: fast movement of funds after receipt, repeated transfers through related accounts, unusually structured amounts, new counterparties receiving many payments, flows that do not match a customer's profile, or cryptocurrency behavior involving anonymity services and risky geography. Regulatory and typology material from FATF, FinCEN, and FFIEC describes this kind of pattern-based reasoning in operational language [Financial Action Task Force, 2020, Financial Crimes Enforcement Network, 2026, Federal Financial Institutions Examination Council, 2026].

This makes the scientific bar different from a normal leaderboard setting. A model score is not enough. A reader needs to know which data stayed local, which columns were allowed at decision time, how time or graph boundaries were split, whether candidate selection touched the test set, what review budget was assumed, and which interpretation the evidence can support. Those questions become sharper when large language models (LLMs) or coding agents assist the research workflow, because fluent explanations can drift away from the artifact record unless the system is designed to fail closed.

Relaytic began as a general local-first inference-engineering lab. Relaytic-AML is the focused financial-crime edition used here to test whether that architecture can support governed AML experimentation. The system is agent-assisted, but artifacts are the authority: specialist roles inspect data posture, propose task contracts, challenge modeling choices, execute bounded runs, review traces, and block unsupported claims through local JSON and paper-generation artifacts. External agents can consume redacted context through the command-line interface, project skills, OpenClaw-style handoff, Claude/Codex skill files, or Model Context Protocol (MCP) adapters, while raw rows and licensed benchmark files remain local unless the operator changes policy.

This paper is therefore a systems and methodology paper. The benchmark rows matter because they exercise the architecture under temporal, graph, operational, and claim-governance pressure. They are not presented as a new best detector family. The central question is whether a local evaluation lab can keep evidence, agent assistance, privacy boundaries, and publishable claims aligned.

The work is organized around four research questions:

- **RQ1:** Can Relaytic-AML produce reproducible evidence cells for AML-style temporal and graph tasks?
- **RQ2:** Can it prevent leakage-prone or unsupported claims from being promoted?
- **RQ3:** Can the local-first artifact graph support rowless handoff to external agents while preserving provenance?
- **RQ4:** Do the benchmark rows demonstrate useful, bounded detector evidence under explicit split and budget contracts?

This paper makes four contributions. First, it presents a local-first artifact graph and role-scoped agent runtime for AML evaluation. Second, it defines an evidence-cell schema that ties each metric to dataset, split, command, artifact, budget, leakage posture, operating point, and claim state. Third, it contributes a release and claim-gating harness that converts local artifacts into tables, figures, source packages, and manuscript claims while blocking unsupported interpretations. Fourth, it gives public benchmark demonstrations on PaySim, Elliptic, and Elliptic2 context rows under strict claim boundaries.

2 Related Work

The AML benchmark landscape is moving toward larger, more realistic, and more graph-native settings. PaySim remains useful as a synthetic mobile-money fraud simulator for temporal proxy experiments [Lopez-Rojas et al., 2016]. Elliptic introduced a public Bitcoin transaction graph with anonymized node features and temporal labels [Weber et al., 2019]. Elliptic2 and RevTrack/RevClassify shifted attention to suspicious subgraphs and richer blockchain context [Bellei et al., 2024, Song et al., 2024]. Recent work such as TransXion, LineMVGNN, quasi-temporal graph extraction, BlazingAML, and continual graph-learning studies shows that the research frontier increasingly treats AML as dynamic graph and systems work rather than static tabular classification [Chen et al., 2026, Poon et al., 2026, Tariq and Hassani, 2026, Ye et al., 2026, Deprez et al., 2025].

Relaytic-AML does not compete with those papers as a new graph-neural detector. Its position is complementary. It asks how an experiment should be governed when the data is local or licensed, the task may be temporal or graph-based, the review queue matters, and an agent-assisted workflow must still produce evidence that a skeptical reviewer can audit. That places the system closer to dataset documentation, model reporting, and reproducibility work [Gebru et al., 2021, Mitchell et al., 2019, Pineau et al., 2021] than to a single detector paper.

The agentic part of the contribution is also important. Recent agent-evaluation work shows that language-model systems can produce persuasive research artifacts while still failing on reproduction, validation, or expert-level judgment [Chen et al., 2025, Starace et al., 2025, Wijk et al., 2025]. Skill- and tool-using agents make that opportunity larger and the governance problem sharper [Yang et al., 2026]. Relaytic-AML responds by making every model score, public claim, handoff packet, and paper asset downstream of local artifacts rather than downstream of a conversation transcript.

3 System Overview

Relaytic-AML is a local evaluation lab built around a simple authority rule: the workspace owns the truth. Raw data, licensed benchmark files, run summaries, traces, metric cells, model outputs, tables, figures, and release reports live on disk. Agents may explain or propose actions, but their proposals only become evidence when they are materialized as artifacts that another human or agent can inspect.

The runtime is organized as cooperating specialist loops. A guide answers where the run is and which artifact matters next. A scout checks source posture, schema, leakage risk, and split feasibility. A strategist turns the user objective into an executable task contract. A scientist challenges the modeling plan and asks for baselines, ablations, or harder budgets. A builder executes bounded benchmark or modeling work. Trace reviewers reconstruct decisions and branch state. Release governors lint paper claims, figure captions, public wording, and source packages against the evidence record. The same loop can be used by a human in the terminal, an external coding agent, or a local LLM-backed guide; the local artifact graph remains the handoff layer.

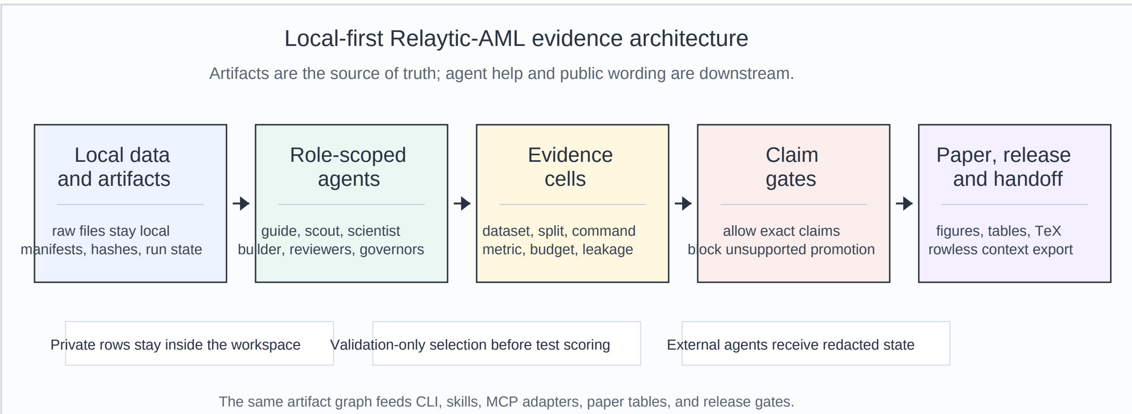

Figure 1: Relaytic-AML local-first architecture: local data and artifacts flow through role-scoped agents into evidence cells, claim gates, and paper/release/handoff surfaces.

Figure 1 summarizes the architecture. Local data and local artifacts flow into role-scoped agents. Those agents produce evidence cells. Claim gates then decide which interpretations can reach the paper, release notes, or a redacted handoff package. This is why the system can be useful even when a benchmark row is only supporting evidence: the row is still connected to the split rule, operating point, leakage posture, and claim boundary.

The external-agent path is intentionally rowless by default. Relaytic can export the current state, action options, artifact shortlist, and safe commands for another model without sending raw transaction rows or private local paths. OpenClaw, Claude/Codex skills, CLI JSON calls, and MCP surfaces all sit on the same contract: external intelligence may help navigate and repair the run, but private data remains local and artifact provenance remains explicit.

4 Evidence Cell and Claim-Gate Design

An evidence cell is the unit that makes a paper number auditable. It is not just a metric value. It records the dataset, split, command, artifact field, model or feature budget, leakage posture, operating point, and claim state. That structure gives a reviewer two kinds of visibility: how the number was produced, and what the number is allowed to mean.

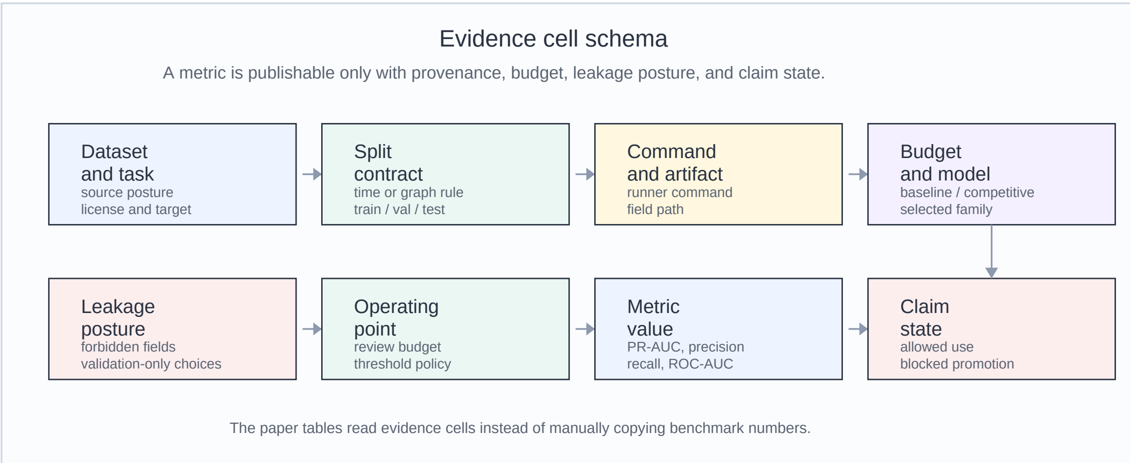

Figure 2: Evidence-cell schema: every reported number carries dataset, split, command, artifact, budget, leakage posture, operating point, metric, and claim state.

Table 1. Representative evidence cells.

Dataset	Metric	Value	Split	Evidence	Budget	Claim state
PaySim	test PR-AUC	0.638773	test	PaySim run; manifest	competitive	supporting only
PaySim	precision at review budget	0.703336	test	PaySim run; manifest	competitive	supporting only
Elliptic	test PR-AUC	0.668756	test	graph run; feature table	competitive	supporting only
Elliptic	precision at review budget	1	test	graph run; feature table	competitive	supporting only
Elliptic2	official test PR-AUC mean	0.94324	official test	E2 run; scorecard	competitive context	context only; no contribution
Elliptic2	published reference PR-AUC	0.974	reported reference	E2 run; scorecard	competitive context	context only; no contribution

Algorithm 1 shows the evidence-cell loop in plain language.

```
for each benchmark task:
    freeze dataset access, license posture, and split contract
    run baseline candidates under the allowed feature policy
    run stronger candidates only within the declared budget
    choose model, calibration, and threshold on validation evidence
    evaluate the selected row once on the fixed test surface
    write metric, provenance, leakage posture, and claim state
```

The claim gate is the second half of the design. It is deliberately conservative: if the evidence cell is incomplete, if a split is leakage-prone, if a metric is only a proxy, or if a stronger interpretation needs a different dataset or study, the gate preserves the evidence but blocks the stronger claim. This is not a cosmetic disclaimer. It is a mechanism that changes what the paper generator and public release surfaces are allowed to say.

Algorithm 2 shows the gate behavior.

```
for each proposed public claim:
    collect the evidence cells named by the claim
    require dataset, split, command, artifact, budget, and leakage provenance
    compare claim strength with source posture and the publishability gate
    allow only the exact interpretation supported by the evidence
    otherwise keep the evidence visible and emit the evidence still needed
```

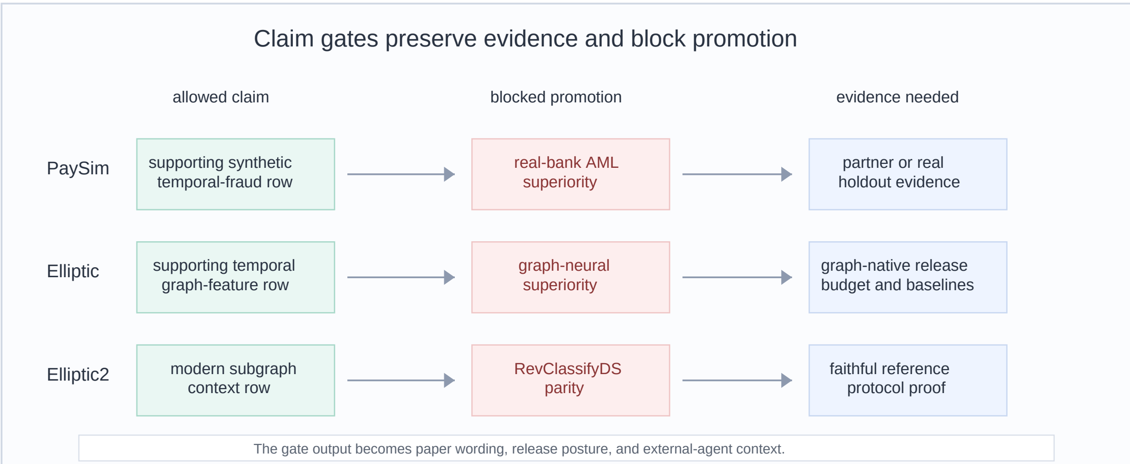

Figure 3: Claim-gate examples: allowed claims, blocked promotions, and evidence needed before stronger public interpretations.

Figure 3 gives the same idea as concrete examples. A PaySim row can support a synthetic temporal-fraud result, but not a real-bank AML superiority statement. An Elliptic row can support temporal graph-feature evidence, but not graph-neural dominance. An Elliptic2 row can be modern benchmark context while the reference-parity claim remains blocked.

5 Experimental Protocol

The experiments are designed to test Relaytic-AML as an evaluation lab. The datasets are intentionally separated by claim posture. PaySim is the main empirical demonstration because it is fully local and supports a controlled temporal proxy workflow. Elliptic tests temporal graph provenance and feature-view discipline. Elliptic2 tests whether the system can keep modern-context evidence visible without promoting it to a performance contribution.

Table 2. Dataset and task contracts.

Dataset	Task	Scale	Split sizes	Split rule	Feature policy	Metrics
PaySim synthetic mobile-money transaction fraud	Synthetic temporal transaction fraud	6,362,620 rows/nodes; 8213 positives	train: 6,010,937, pos 0.08%; validation: 228,103, pos 0.68%; test: 123,580, pos 1.34%	time step	26 decision-time features; balance columns excluded	PR-AUC, P@k, R@budget
Elliptic Bitcoin transaction graph	Temporal Bitcoin graph node classification	203,769 rows/nodes; 4545 positives	train: 26,381, pos 10.88%; validation: 8,999, pos 11.53%; test: 11,184, pos 5.69%	graph time	structural same snapshot only; source provided flattened features; source features plus structural snapshot	PR-AUC, P@k, fixed-FPR recall
Elliptic2 large Bitcoin subgraph AML dataset	Suspicious-versus-licit subgraph context	122,000 labeled subgraphs; context row	val 11,059, test 11,105; test positives 272	fixed partition	pooled moments counts; raw subgraphs local-gated	PR-AUC, P@k, fixed-FPR recall

Table 2 records the task contracts that a reader needs before interpreting any metric. The positive rates show why precision-recall area under the curve (PR-AUC) is the primary score: these are rare-event tasks where receiver-operating-characteristic area under the curve can look strong while the review queue is still poor. The split column is equally important. PaySim is split by chronological simulator step, Elliptic by time-step graph windows, and Elliptic2 by the official/context partitions recorded in the local evidence pack.

Table 3. Model families and search budgets.

Track	Model families	Feature families	Search budget	Selection rule	Seeds
PaySim	Extra Trees, Random Forest, HistGradient-Boosting, LightGBM, XGBoost, fixed rule	amount, type, time, and prior-step destination history	14 probes; 5 full finalists	validation PR-AUC only; Platt calibration	42
Elliptic	Extra Trees, HistGradient-Boosting, LightGBM, XGBoost	source anonymized, structural snapshot, and combined graph features	16 validation trials; LightGBM selected	validation PR-AUC within declared feature views; validation-only calibration and threshold	42
Elliptic2	Pooled-moment LightGBM context candidate	pooled subgraph moments/counts	3 repeated seeds; context row only	official partition plus content-hash robustness check; reference parity gate remains blocked	11, 42, 73

Table 3 records the modeling effort. PaySim uses a two-stage competitive budget: probes over a seeded train-only sample, then five full-training finalists, validation-only calibration, and one fixed test evaluation for the selected finalist. Elliptic uses a graph-feature budget over source-provided anonymized features, same-snapshot structural features, and their combination. Elliptic2 uses repeated pooled-moment LightGBM context rows with seeds 11, 42, and 73, but this row is not a Relaytic performance contribution because reference-protocol parity is unresolved.

The feature policy is strictest for PaySim because simulator balance columns can leak post-event state. Relaytic excludes those balance fields, raw origin and destination identifiers as model features, and simulator flags. It allows row-local amount, type, and time features, train-only thresholds, and destination history shifted before each step. For Elliptic, source-provided anonymized node features are kept separate from Relaytic-derived label-free structural graph features so the claim cannot blur source features into a graph-engineering win.

6 Results

Table 4. PaySim modeling path and ablation evidence.

Stage	Model/contract	Change	Test PR-AUC	Claim use
P4 reference	SGD logistic baseline	source safe starting point	0.215852	not a paper claim
P6 baseline	Extra Trees baseline	leakage-safe feature set	0.331345	baseline only
Destination history	destination history contract	balance fields excluded; shifted destination aggregates added	pending isolated test	feature contract proved, isolated ablation missing
Competitive search	XGBoost probe	best validation PR-AUC 0.594444	test hidden until selected finalist	search evidence only
Final model	Extra Trees with Platt calibration	validation-selected finalist evaluated once on test	0.638773	supporting synthetic temporal-proxy claim

The isolated destination-history cell is intentionally marked pending: `TODO_EVIDENCE[paysim_prior_history_isolated_test_pr_auc]`.

PaySim is the clearest modeling result in the current paper. The earliest reference row was weak, with PR-AUC 0.215852. The later leakage-safe baseline improved to 0.331345, and the competitive selected Extra Trees model reached fixed-test PR-AUC 0.638773 and ROC-AUC 0.968282. The improvement is meaningful inside the synthetic temporal-fraud contract because it comes from a visible sequence of choices: forbidden balance fields were excluded, prior-step destination history was added without raw account encoding, candidates were selected by validation PR-AUC, and calibration and thresholding used validation-only partitions. The allowed claim is therefore precise: Relaytic-AML can produce a stronger, leakage-audited PaySim temporal-proxy row under a declared budget. It cannot turn that row into evidence of real-bank AML superiority.

The review-budget metrics sharpen the interpretation. At the selected PaySim review budget, precision is 0.703336 and recall is 0.471584. This means the top of the queue is much richer than prevalence, but a large share of fraud remains outside the reviewed set. That is a useful operating result for an evaluation lab because it connects model ranking to analyst capacity instead of treating PR-AUC as the whole story.

Elliptic is a different kind of evidence. The validation-selected source-plus-structural LightGBM row reports test PR-AUC 0.668756, with review-budget precision 1 and recall 0.056604. The result supports temporal graph-feature provenance and operating-point reporting. It also reveals a limitation: the current graph-structure-only floor is weak, and the final row is heavily influenced by source-provided anonymized features. Relaytic can therefore claim graph-aware evidence handling, not a new graph-native detector.

Elliptic2 is intentionally framed as context. The repeated official-partition context row reports PR-AUC 0.94324 +/- 0.000882, and the content-hash robustness partition reports mean PR-AUC 0.929669. Those are strong absolute values, but the recorded RevClassifyDS reference is 0.974, and the local parity gate records unresolved cohort and reference-execution issues. The scientific result is that Relaytic keeps the context row available while blocking a parity or superiority interpretation.

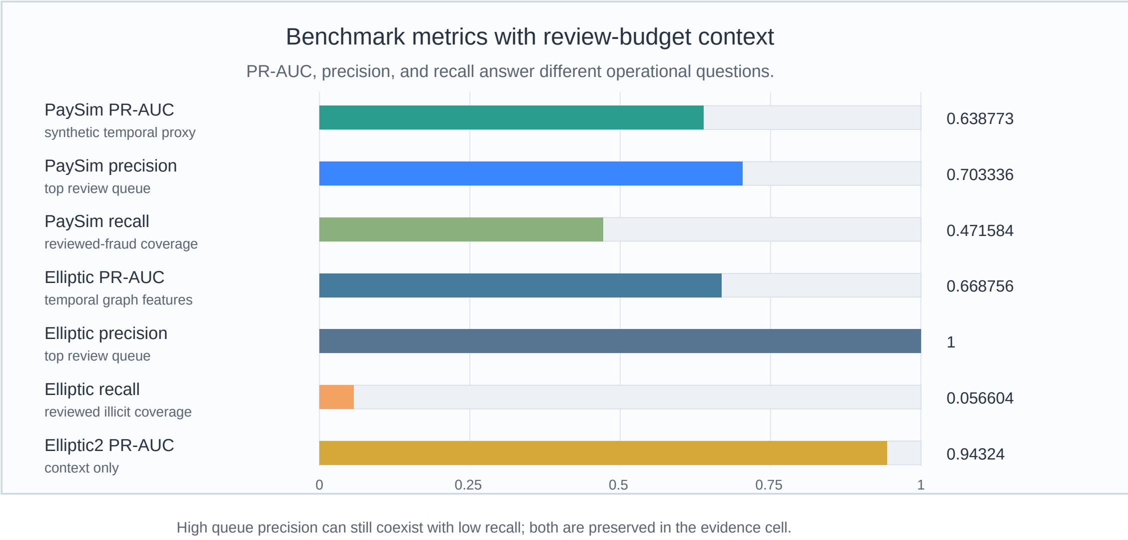

Figure 4: Benchmark and review-budget evidence: PR-AUC is shown beside precision and recall at the bounded review queue instead of being interpreted alone.

Figure 4 places the benchmark metrics beside review-budget behavior. The visual point is not that one bar wins everything. The point is that Relaytic keeps ranking metrics and operational metrics in the same evidence surface, so a reader sees both model quality and review-queue consequences.

7 System Evaluation

The system claim is evaluated through deterministic reader and agent tasks. These checks ask whether a reviewer can navigate from the README to the paper evidence, whether PaySim metric provenance is recoverable, whether baseline and competitive budgets are comparable, whether Elliptic2 remains visible as context rather than a performance contribution, whether rowless external-agent handoff works, and whether claim gates fail closed. This is narrower than a human-subject usability study, but it directly tests the paper's infrastructure claim.

Table 5. Deterministic system-evaluation protocol.

Audit axis	Protocol	Evidence surface	Outcome
Navigation and scope	README separates the general Relaytic lab from the focused AML paper path.	reader navigation check	documented
Metric lineage	PaySim result rows resolve to dataset, split, command, artifact, and claim posture.	metric-cell audit	traceable
Model-search budget	Baseline and competitive PaySim rows are compared under declared budgets.	budget audit	comparable
Claim governance	PaySim, Elliptic2, hard claims, and headline claims stay inside their evidence boundary.	publishability matrix	fail-closed
External-agent handoff	A coding agent or local model receives redacted state, tool choices, and next actions without raw rows.	handoff evaluation	rowless
Interrupted-run recovery	A new or interrupted user can see current state, missing evidence, and canonical next artifacts.	recovery evaluation	recoverable
Cross-platform rerun	Windows and macOS/Linux commands regenerate the paper and system-evaluation artifacts.	README command path	documented

Table 5 summarizes the protocol audit behind the system claim. The important result is not that a script returned `pass`; it is that the reader-facing behaviors are recoverable from local artifacts: metric lineage is traceable, model-search budgets are comparable, claim gates fail closed, and external-agent handoff can be redacted. The current deterministic suite reports no raw-row exposure and no private-path exposure. That supports RQ2 and RQ3, while leaving true human usability and analyst-hour validation for future studies.

8 Limitations and Threats to Validity

The current evidence has real limits. PaySim is synthetic mobile-money fraud data, so it is useful for temporal proxy evaluation but cannot establish real-bank AML performance. Elliptic is public blockchain data with anonymized features and unknown-label scope, not a bank transaction environment. Elliptic2 is a modern subgraph benchmark, but the local context row is not a faithful RevClassify reproduction and the recorded official test exposure means the result must stay context-only.

There are also evaluation threats. Deterministic system checks are not the same as controlled human usability studies. Review-budget rows estimate queue behavior under a fixed public benchmark surface; they do not prove analyst-hour savings, case quality, or production review performance against an incumbent system. The public tasks may encourage overfitting to known benchmark protocols, and future work should test stronger release budgets, repeated runs, private or partner-approved holdouts, same-queue incumbent comparisons, and graph-native families under the same evidence-cell discipline.

The system is also intentionally local-first, which creates a tradeoff. Privacy and provenance improve because raw rows stay local, but external reviewers cannot rerun licensed or private data without obtaining it themselves. The paper handles that by publishing commands, hashes where allowed, generated artifacts, and claim boundaries, but a fully independent reproduction of every heavy benchmark still depends on legal access to the source datasets.

9 Reproducibility

The repository is larger than this AML paper. Relaytic is the general local-first inference lab and public package; Relaytic-AML is the focused AML edition used here for the manuscript. A reader should start with the README and this paper. Development-control files record the build history, but they are not required to understand the paper claims.

Minimal regeneration commands are shown below.

Windows PowerShell:

```
py -3.11 -m pip install -e ".[full]"
py -3.11 -m relaytic.ui.cli release-safety paper-system-eval --format json
py -3.11 -m relaytic.ui.cli release-safety paper-release --format json
py -3.11 -m relaytic.ui.cli release-safety paper-arxiv-source --format json
py -3.11 -m pytest ‘
  tests/test_paper_track_p13.py ‘
  tests/test_paper_track_p14.py ‘
  tests/test_paper_track_p15.py -q
```

macOS/Linux:

```
python3 -m pip install -e ".[full]"
python3 -m relaytic.ui.cli release-safety paper-system-eval --format json
python3 -m relaytic.ui.cli release-safety paper-release --format json
python3 -m relaytic.ui.cli release-safety paper-arxiv-source --format json
python3 -m pytest \
  tests/test_paper_track_p13.py \
  tests/test_paper_track_p14.py \
  tests/test_paper_track_p15.py -q
```

Raw benchmark data is not committed. PaySim and Elliptic require local downloads and are referenced through registry artifacts, split reports, hashes, and command ledgers. Elliptic2 remains context-only in this paper because the stronger reference-parity conditions are not satisfied locally. Final arXiv upload still requires `TODO_EVIDENCE[author_metadata]`, human PDF inspection, and a clean tag target.

LLM tools assisted with drafting, editing, repository inspection, and consistency checks. They are not authors. The evidence cells, code, figures, tables, limitations, and final claims remain the author's responsibility.

10 Conclusion

Relaytic-AML shows how an agent-assisted AML evaluation lab can be built around local evidence rather than conversational memory. The system keeps data posture, temporal and graph split validity, leakage controls, model budgets, review-budget operating points, rowless handoff, and public claims inside one artifact graph. The PaySim, Elliptic, and Elliptic2 rows are useful because they demonstrate that architecture under realistic forms of pressure, including rare events, graph provenance, modern benchmark context, and blocked claims.

The strongest claim supported today is architectural: Relaytic-AML can make AML experiments easier to inspect, easier to challenge, safer to hand off to another agent, and harder to overstate. Stronger detector and business-value claims require stronger evidence. That is exactly the boundary the system is designed to preserve.

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